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MATH-014-01: Introduction to Data Science
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Crimes and their Patterns: Visual Inspections

Objective:

This data set, *Crimes in Washington D.C.*, focuses on the frequency of crime across multiple variables. The primary objective of this project is to understand and analyze the different hours, locations, and other social factors and its contribution towards the frequency of crimes occurring in Washington, D.C. Within this study, real-world data is used in order to understand my primary focus: whether there are different frequencies of crime throughout the day.. Additionally, other factors such as geographical location and neighborhood cluster are analyzed to understand other influential variables and their impact. Analyzing this data will help bring awareness towards data-driven mitigation strategies and utilization of resources to prevent such crimes.

Introduction:

This data set focuses primarily on crimes that have occurred in the year of 2025 in Washington, D.C. Including multiple measured variables, its keen variables measured the latitude and longitude (geographical location), in which crimes occurred, shift in which crimes had the highest frequencies, and specific times of when the crime was committed. This data is important because it serves as data focusing on real-world problems that will translate into awareness of patterns within certain environments. Not only is this important for knowledge, but for the general protection of individuals within this specified area. Being aware of specific details such as time and location will assist governing leaders as well as authorities think and plan ahead. The project that I am exhibiting attempts to understand whether or not the frequency of crimes depend on the time of day. My main research question is *How does the frequency of crime fluctuate at various times of the day?*

Methods:

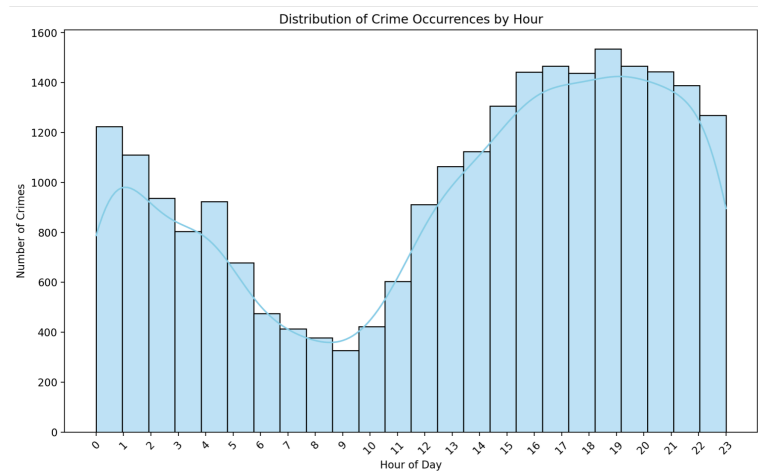
When analyzing this data, it is important to understand what it is displaying before beginning the data cleaning process. The original data contains multiple variables such as report date, the exact start date of the crime, the type of offense committed, the neighborhood cluster, voting precinct, and more. Understanding the different variables and data is important before beginning the cleaning process so that you may drop irrelevant variables towards the research study. The first step I took in beginning the data cleaning process is understanding and presenting the first and last five rows of the data set to understand the different variables and to display certain patterns. I also checked for the rows and columns to have knowledge of how much data

this set includes. After doing so, I found that the crime data set included 24,151 rows and 25 columns. This is an exceedingly large data set, so understanding this before beginning the cleaning process is extremely important. Other necessary steps were taken to better understand the data I am working with. Steps taken include checking for data types, and looking at the summary statistics. After understanding the data, I took the necessary and mandatory steps to clean it. The first step I took is checking for missing values. There were missing values within certain variables; however, these were not necessary towards my research question and further analysis led me to drop these irrelevant columns. Other cleaning measures include checking for duplicated variables to ensure consistent and clean data cleaning process, standardizing column names and string entities, to create a clear and efficient variable list, and encoding the categorical column, offense type, into numeric values in order to look at correlations and graphs that use numerical values. An important factor within the data set that was cleaned includes the conversion from the date variables (start_date) to datetime format. At first, the variable values were listed as a string, which would not work since we are looking at numeric values. Keeping these values would confuse python and not translate it into time because it is a string. Converting this to datetime form will better assist me in analyzing graphs, looking at specific hours, and understanding spikes or patterns within my data set. Once the data was converted, it was essential that I created a value named "hour" which focuses on the specific hour of the day in which a crime occurred. This value was derived from the timestamp within the start_date column. This standardization made a keen and significant difference within the data set, as it made trends and patterns easier to analyze and interpret. This also makes the data analyzation process easier for the researcher, as looking at timestamps may be more confusing and time consuming than looking at a specified hour. In order to better interpret what the data shows, visualizations were made to analyze how crime shifts in regards to time. Additionally, analyzing factors such as neighborhood cluster and geographic location were also completed.

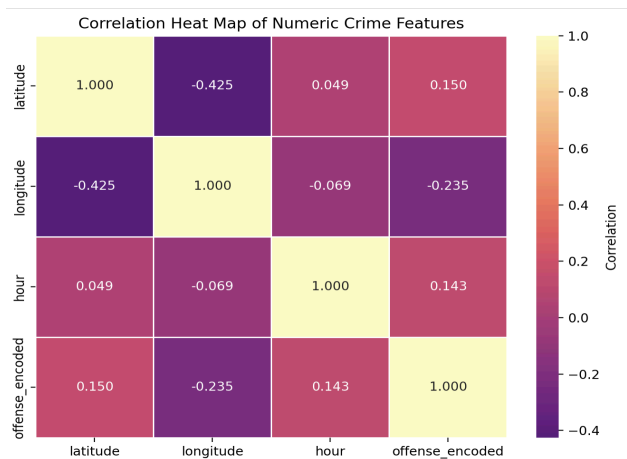
Data Visualization and Interpretation:

The first data visualization graph coded was a univariate graph, the histogram. The histogram analyzes the number of crimes that are committed per hour. As seen directly on the graph, the x axis is the specific hour of the day and the y axis exhibits the number of crimes that occur within that hour. The hours are listed from 0-23, each representing a certain time of the day. Within the graph, the bars exhibit a uniform pattern. The crimes have high variability in regards to hours and are not equally spread. There is a decrease in crimes from the hours of 5-9, suggesting that crime is lowest during the earlier part of the morning. Analyzing the earlier morning and evening data shows that the bars are high, translating into a peak of crimes within those hours. The times that include the highest amount of crimes are hours 15-21 which represent 3-9 p.m. Though crimes still occur after this hour, a slight decrease exists within the data. Although no outliers exist within this chart, skewed data exists in which major differences of crimes are evident across specific times. A key insight retrieved from this visualization is that

crime frequency does differ by hour with its highest rate being during the evening. This supports my research as it affirms that crime frequency does differ by the time of day.

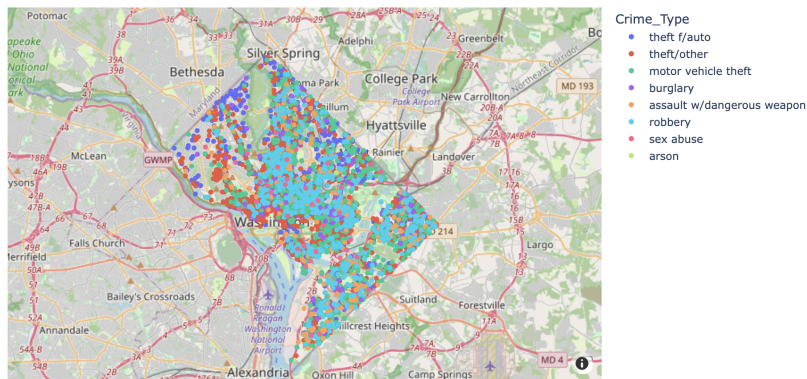


The next visualization, the heatmap, was conducted to understand if there was a direct relationship within my variables. Is this a simple relationship that could be explained by numeric values? As one value increases, does the other increase as well? I wanted to analyze and show if there was a specific trend that the variables follow with one another. This graph analyzed the strength and relationship between a specific offense type with latitude/longitude (geographic location), and hour. Within each box lies the relationship between the two variables. This graph shows all values to be close to zero, indicating no strong-linear relationship whatsoever. An example is the relationship between hour and offense type. The value is 0.143 which shows little to no relationship. This affirms the notion that the variables within my study do not hold a simple linear relationship. As time increases, crimes and their types do not increase or decrease. The occurrence of crimes and their times are explained by multifaceted patterns that hold spikes rather than following a specific linear path. Rather than focusing on numeric variables, it is shown by the heatmap that crime is influenced more by qualitative distributions.



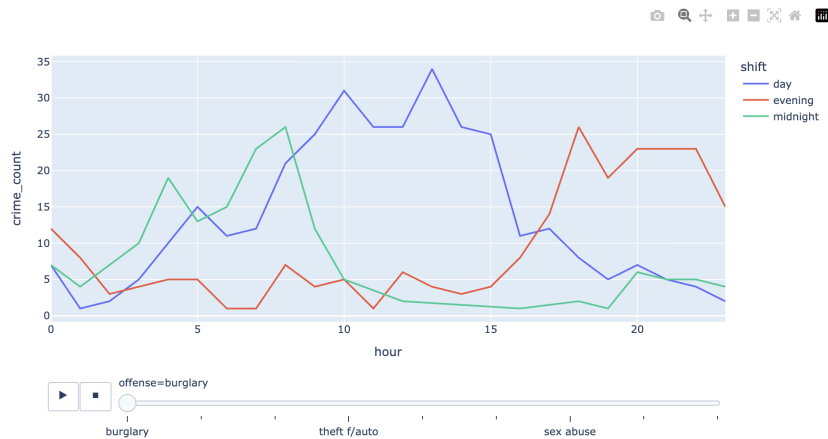
The next graph used to analyze this data is the geo-map, an interactive graph that displays the location based dispersion of crime incidents that occur within the Washington, D.C area. On the right side of the geomap, the colored points represent the specific crime type. On the map, the cluster of points represents the crime frequency and how often they occur. A significant point that I analyzed is the pattern of clustered points in the center of the map. The area in which these points are clustered is downtown/central D.C. Specific areas in this area include Dupont Circle, the National Mall, Chinatown, and other heavily populated areas. A pattern I noticed was the crime frequency of the specific crime type, armed robbery. This makes sense, as these specific areas include many shopping centers, high tourist attractions, and a deep population. Although there are no significant outliers, analyzing the map further shows a small amount of points outside the main region. These areas represent the unusual areas outside of the main geographical region. A key insight is that situational factors, such as the sociological environment influence where crimes occur.

Interactive Map of Evening Crime Incidents

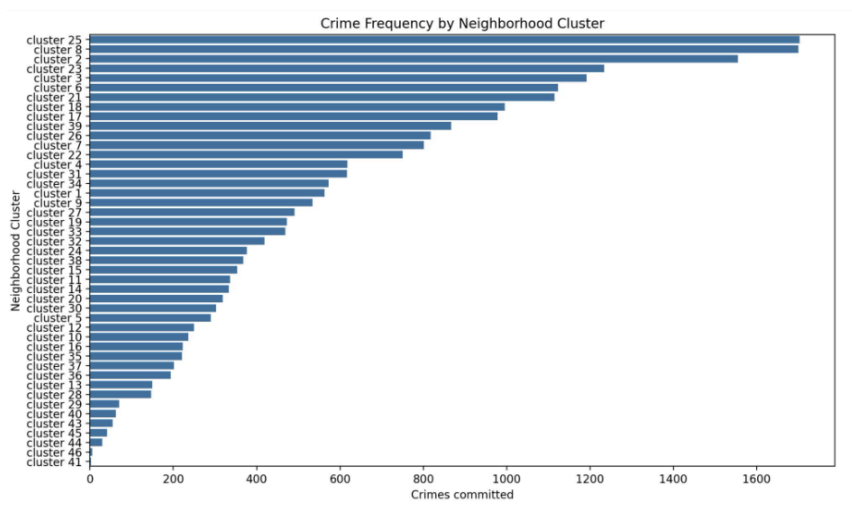


The next graph I used to visualize my data is an animated line-plot chart, which displays the difference in crime occurrence amongst specific hours of a shift and the type of crime committed. On the x-axis lies the specific hour in which crimes occur while the y axis is the crime count. The three colored lines represent the specific shift in which the crimes occur. The dark blue line represents day, the orange represents evening, and the light blue represents midnight. When animating the graph, it shows that the evening line has the highest spikes throughout the crimes and hours. The hours in which crimes spike the highest within the evening shift is hours 15-21, as seen within the other visualizations. When examining the day line, the spikes are not as high and exhibit a more gradual shift. There seems to be no outliers that exist within this visualization; however, the patterns show that there are steady spikes that occur within evening crimes despite the type of crime. Important insights gained from this visualization

reaffirm my research by showing that the time of day does influence crime occurrence despite the offense type.

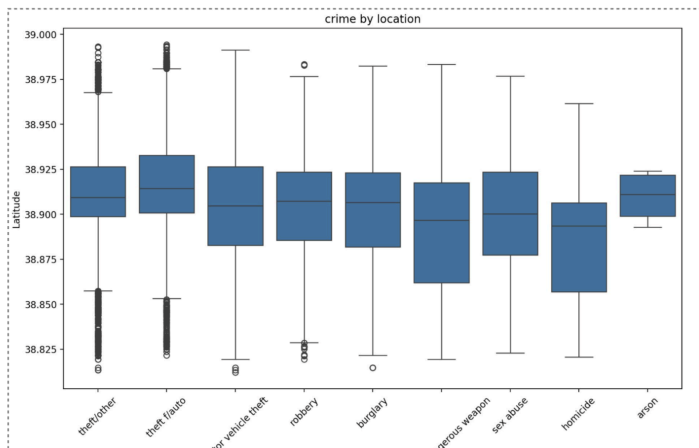


The next graph I chose as a visualization of my data focuses on additional analyses outside of my main research question. This graph will examine if crime *frequency differs by neighborhood cluster*. The type of graph I chose is a countplot which uses bars to show the amount of crimes that occur within the specific neighborhood. This graph shows a significantly different frequency of crimes between the neighborhoods. For example, clusters 2, 8, and 25 have more crimes while clusters 41, 46, and 44 and less than 200 crimes that have occurred in 2025. This graph supports my analysis of the question that crime frequency does differ amongst neighborhoods. It is evident that neighborhoods hold different laws and regulations as well as other social factors that account for this difference.



The final graph I chose focuses on another additional analysis question: *Does latitude (geographic location) vary by the type of offense committed?* The visualization used for this

question was a box plot. On the x-axis lies the specific crime that was committed, while the y-axis displays the latitude or specific geographic location. It is important to note that the values listed for latitude, ranging from 38.825-39.000, are located in downtown/central D.C. It is interesting to see that the median, (the white line that lies within the interquartile range), lies between similar points across similar geographical values (38.89-38.91). The small dots outside of the average value represent outliers, crimes that occur in unusual locations from the median. The only crimes that had existing outliers include theft/other, theft/auto, vehicle theft, robbery, and burglary. An insight gained from this visualization is that the majority of crimes and their crime types do occur in the same area with some crimes existing outside of the regular locations.



Limitations & Real World Solutions:

Though this data set contained many variables that were necessary to be measured, it would have benefited categorical values such as economic and social classification, laws and police enforcement to be added to understand the reasoning behind certain data points. Another limitation is the fact that there are many crimes that may have not been reported within the Washington, D.C area, meaning the statistics may be missing important values. In regards to real world solutions, data-driven initiatives are extremely important so that less crimes could occur. When looking at this data, authorities should implement initiatives that will focus on the specific times and neighborhood clusters where crime is at its highest. For example, increasing surveillance or even police within the heavily populated areas and during the high-crime hours.

Conclusion:

In conclusion, my research question was affirmed by showing that different hours of the day have different crime frequencies. The time in which crime occurrences reach its highest point is during hours 15-21, which represent the time of 3-9 p.m. In addition, the first analysis question that asked if crime frequency differed by neighborhood cluster was confirmed, showing that neighborhood clusters do significantly differ from one another, suggesting influence of sociological factors and differences within neighborhood laws, regulations, and security measures. The data and visualizations also show that the majority of crimes happen within the

same geographical region. Another significant insight that is taken from this study is that crime does not hold a linear relationship with variables such as hour or geographic location, but is rather influenced by other sociological and qualitative factors that influence the spike in certain hours.

References

District of Columbia Open Data. (2025). *Crime incidents in 2025*. Washington, D.C. Government. <https://opendata.dc.gov/>